EMBER

AI-Based Wildfire Occurrence Risk Assessment

Unit 5 / Project 3 / Bayesian Model Averaging comparative study

1. Objective and honest target

Target: 1[recorded burned area > 0], conditional on a recorded fire. Area = 0 is not a confirmed non-fire observation. This dataset cannot validate daily ignition risk.

Compare probabilistic environmental models and assess calibration and uncertainty. The implemented target is a documented proxy, not a claim of operational fire occurrence prediction.

2. Data and experiment protocol

Nine scenario inputs become ten predictors: temperature, RH, wind, log1p(rain), DMC, DC, FFMC, ISI and cyclical month sine/cosine. Coordinates only define groups; area only defines the target; weekday is excluded. Invalid or nonfinite inputs fail validation.

Use whole (X,Y) spatial cells: train/validation/test seeds 42 and 43, unchanged from the original repository. Training-only four-fold GroupKFold minimizes Brier score; scalers fit inside each fold. Validation selects F1 thresholds; no test-fitted calibration. All algorithms share identical validation and test rows.

Split	Records	Cells	Positive fraction
train	237	21	0.464
validation	89	7	0.539
test	191	8	0.586

3. Primary algorithm and tuning

Enumerate all 16 subsets of weather, moisture, fire-index and season blocks. Each includes an intercept. Coefficients have normalized Gaussian priors; independent Bernoulli block priors control model complexity. Laplace marginal likelihood times model prior yields posterior weights. Twenty-point Gauss-Hermite quadrature integrates coefficient uncertainty in predictions.

Expanded training-only grids: BMA prior SD {0.15, 0.35, 0.7, 1, 2.5}, block inclusion {0.2, 0.5, 0.8}; Logistic Regression C {0.001, 0.01, 0.1, 1, 10, 100}; Decision Tree depth {1,2,4,6}, leaf size {5,15,30}, pruning alpha {0,0.01}; Random Forest 300 trees, depth {3,unrestricted}, leaf size {5,15,30}, max_features {0.5,1}.

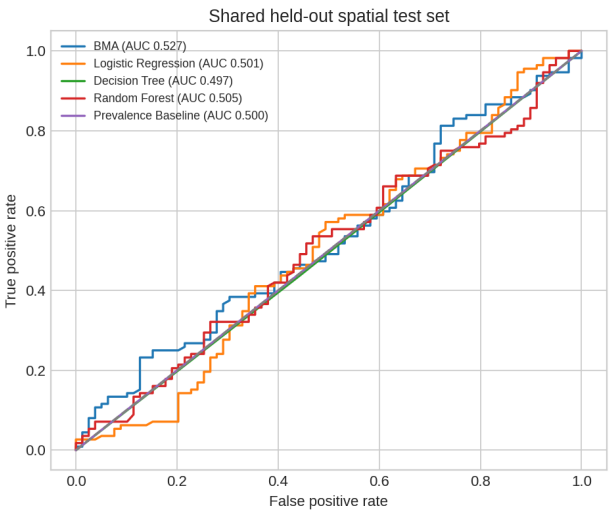
4. Measured comparative results

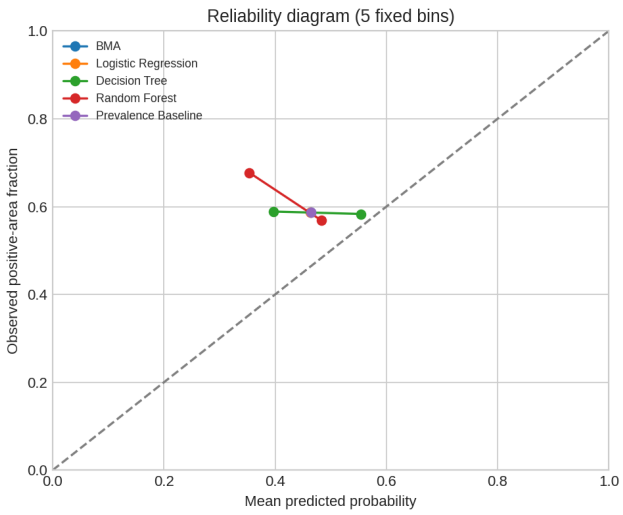
Algorithm	Val. Brier	Test AUC	Test F1	Test Brier
BMA	0.2540	0.5267	0.7393	0.2574
Logistic Regression	0.2523	0.5007	0.6561	0.2577
Decision Tree	0.2257	0.4972	0.5000	0.2635
Random Forest	0.2370	0.5054	0.6590	0.2620
Prevalence Baseline	0.2541	0.5000	0.7393	0.2575

BMA test ROC-AUC is 0.5267, F1 0.7393, Brier 0.2574, balanced accuracy 0.5000. Baseline F1 is also 0.7393. BMA predicts every test record positive at the validation-selected threshold; this F1 is not evidence of useful discrimination. Expanded tuning does not establish a meaningful improvement in test performance.

Higher ROC-AUC and F1 are better; lower Brier is better. Validation F1 is optimistic after threshold selection. Training times include the complete search; inference times average 30 warmed batch predictions divided by row count, excluding I/O.

Algorithm	Train/search (s)	Inference (ms/row)	Interpretability
BMA	2.108	0.0215	Weighted logistic subsets; block inclusion probabilities
Logistic Regression	0.227	0.0090	Standardized coefficients
Decision Tree	0.780	0.0152	Decision rules (depth-limited)
Random Forest	25.187	0.1131	Ensemble; global feature importance
Prevalence Baseline	0.029	0.0074	Training positive-class frequency

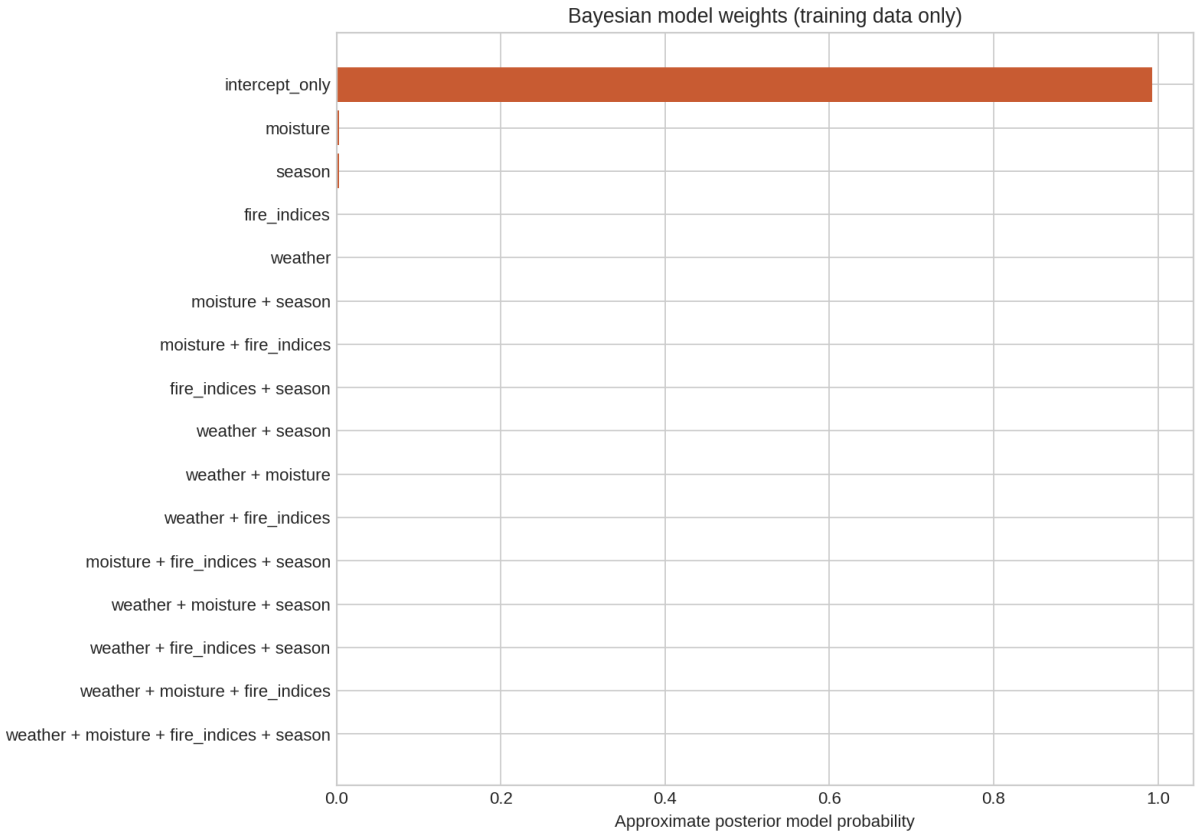




5. Bayesian evidence and uncertainty

Most supported subset: intercept_only (weight 99.33%). Weights sum to one and are conditional on this family, preprocessing and selected priors.

Pointwise 90% BMA latent-probability credible intervals sample the model index and Laplace coefficient posterior (2,000 draws, seed 42). Conditional on the candidate family; they do not account for dataset bias. Test metric 95% percentile intervals resample spatial cells 500 times (seed 2026), skip one-class resamples, and do not refit models. They describe conditional sampling variability, not external validity.



BMA metric	Lower 95%	Upper 95%
roc_auc	0.4405	0.6162
f1	0.7140	0.7685
brier	0.2552	0.2601

6. Application and submission checklist

The Streamlit interface offers an editable scenario, all model probabilities, Bayesian credible intervals, training-range warnings, JSON export, batch CSV scoring, single-input sensitivity, calibration and ROC plots, model weights, dataset exploration and downloads.

PDF requirement	Delivered evidence
UCI data / careful target / weather	data/forestfires.csv; data.py; notebook sections 1-3
Primary Bayesian Model Averaging	bma.py; all 16 model weights and block-inclusion plot
LR / Tree / Random Forest comparisons	model_comparison.csv; *_cv.csv; shared test rows
ROC-AUC / F1 / Brier / calibration	comparison table; roc_calibration.png; calibration_bins.csv
Probability interface / weights	app.py: scenario studio, Bayesian evidence, batch scoring
Notebook / repository / report / slides	Executed notebook; GitHub; this PDF; editable PPTX
Timing / interpretability / audit plots	Full comparison CSV; split assignments; predictions; plots

7. Limits and recommended next steps

Only 517 records from one Portuguese park, with no representative non-fire controls. Month/weekday are not complete timestamps. Spatial grouping reduces location leakage but does not remove all event/weather dependence. Test includes only eight cells. This revision reuses previously published test scores; it is not fresh external validation. Do not use this educational application for emergency decisions.

Obtain representative non-fire observations and exact dates, prespecify the observation window, then validate on genuinely new regions and time periods. Better data, not optimistic splitting, is needed for credible operational performance. Student name, roll number and institution should be supplied by the student; none have been invented in these deliverables.

Reproduce and reference

python -m wildfire.train; python -m wildfire.deliverables; python -m pytest -q; python -m streamlit run app.py

Cortez, P. & Morais, A. (2007). Forest Fires. UCI Machine Learning Repository. https://doi.org/10.24432/C5D88D (CC BY 4.0). Code: https://github.com/RomitDeokar/wildfire_risk_ml

Python 3.13.14; SHA-256: 0d6586a1fa52f55bef48578aef14eb97273f1e9330e1a53423df497a77065253